Marco Del Fava

Research Software & Neural Data
M.S. Biomedical Engineering student | UC Davis | Human intracranial electrophysiology
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Research Profile

Biomedical engineering graduate student at UC Davis developing Python software for human intracranial electrophysiology. My work includes adapting readers for proprietary recording formats, repairing acquisition metadata, building neural processing pipelines, and validating data integrity. I use coding agents extensively, with profiling, debugging, and automated tests to evaluate their implementations.

Technical Skills

Scientific Programming: Python, NumPy, Pandas, scikit-learn, pytest, Git, Linux, Docker; MATLAB

Neural Data & Acquisition: Neuralynx, Intan RHD, DarkHorse MED, Neo, SpikeInterface, Kilosort4, MountainSort5

Engineering Methods: Scientific data parsing, metadata validation and repair, waveform analysis, experimental synchronization, profiling, regression tests

Research Experience

Research Volunteer, UC Davis Health — Z-Lab (Dr. Jie Zheng) — Jul 2025 - Present

- Work with human intracranial electrophysiology recordings, developing Python workflows to prepare, process, and analyze single-neuron data.
- Develop event-aligned analyses of responses to repeated spoken narratives, combining sorted spike timestamps, experimental markers, rasters, and peri-stimulus firing rates.
- Support acquisition and spike-processing workflows across recording systems, including channel mapping, signal quality assessment, and synchronization with behavioral experiments.

Selected Engineering Projects

Neural Data Formats and Integrity

- Adapted a Neo-backed MED reader to skip corrupt parity channels whose metadata caused standard readers to abort, allowing processing without modifying source recordings.
- Developed validation and repair utilities for Neuralynx recordings with sample-count inconsistencies and missing recording-end metadata that prevented downstream conversion.
- Evaluated signal fidelity across 20 million samples on one channel (32 kHz; 625 s), comparing original CSC and compressed MED recordings. Measured a maximum amplitude difference of 0.5 uV, with exact agreement after whole-microvolt source rounding.

Source

Neural Acquisition and Experimental Synchronization

- Built Python workflows for processing Intan recordings through SpikeInterface, including probe mapping, channel checks, spike sorting, waveform extraction, and quality metrics.
- Developed synchronization software to align PsychoPy behavioral events with RHX recordings, using matched event labels and affine clock fitting to characterize drift between recording systems.

Source

Selected Engineering Projects (continued)

Waveform Analysis and Spike Sorting

- Evaluated waveform representations across 38,145 spike snippets and 10 putative O-Sort unit labels, comparing how well different features preserved local waveform structure and supported clustering.
- Found that strong local neighbor consistency did not translate into equivalent global clustering agreement, distinguishing local waveform-feature consistency from whole-dataset clustering behavior.

Source

Debugging Agent-Generated Software

- Investigated a performance regression in Hosaka, a command-line software platform, where agent-generated schema validation made listing seven records take approximately ten seconds.
- Used execution timing logs to identify repeated validation as the bottleneck, then introduced content-based caching and regression tests to preserve checks for changed or invalid schemas.

Source

Additional Professional Experience

IT Administrator, First Step Communities — Jul 2021 - Present

- Maintain network and IT infrastructure across multiple shelter sites, including routing/VLAN segmentation, Microsoft 365 administration, and troubleshooting.
- Support longitudinal data collection and analysis of service utilization and housing outcomes, translating operational data into research and reporting workflows.

Analyst, Eurofins Test America — Oct 2019 - Jun 2021

- Prepared soil and water samples using solvent extraction in a high-throughput environmental testing laboratory, following established quality assurance procedures.
- Maintained high-accuracy data entry and review in a regulated laboratory environment; supported QA documentation and adherence to standard operating procedures.

Education

M.S. Biomedical Engineering, University of California, Davis — 2026–Present

M.S. Psychology, Arizona State University — 2023

B.S. Neuroscience, Physiology, and Behavior, University of California, Davis — 2017

Professional Development & Certifications

Python for Data Science, Professional Concentration, UC Davis Continuing and Professional Education — 2025

- Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries — UC Davis Continuing and Professional Education (2025-09-18)
- Biomedical Research; Research Administration; GCP Social and Behavioral Research — CITI Program